

technology and enable high-density mounting by laying out components in three dimensions.

Chip SMD inductors are another example of a product designed specifically for power-management applications. Downsizing these is a suitable solution for low-profile and miniaturisation applications within the magnetics market. At a basic level, choosing the right supplementary design materials can also help manage larger footprints without compromising on higher currents.

Polymers and other materials

Polymer and hybrid capacitors offer higher ripple currents, operate at lower equivalent series resistance (ESR) and are more compact, explains Honnavar. Lower ESR reduces AC ripples in power circuits and often reduces peak-to-peak voltage changes by up to five times when compared to tantalum capacitors.

The capacity of polymer capacitors does not drift in response to temperature changes unlike ceramic capacitors that experience instability. Hybrid capacitors also feature stable capacity against conventional liquid-electrolytic capacitors that show a drop in performance.

Vishay has also announced a new series of precision thin-film chip resistors featuring gold terminations for conductive gluing. Offered in 0402 and 0603 case sizes, the MC ATAU series combines high-temperature operation to +155°C with tight tolerances to ± 0.1 per cent.

Caddock Electronics has brought out a series of power-film resistors that come with standard lead finish of solderable thin gold, for applica-

Passive components: Present and future

Today India is importing practically the entire demand of passive components from China, including resistors, capacitors, wire-wound components and much more. Local production of components was hampered by low Indian demand, absence of local raw material supply, poor technological base, lack of skilled manpower, lack of initiatives and interest from entrepreneurs to invest in local manufacturing, inverted customs duty structure, poor infrastructure and unreliable electric supply, among others.

What next? Due to the changed scenario in India, new technologies are coming up. The rise of educated populace has led to the demand for latest technology products. The government's initiative to push for local manufacturing has seen the entry of world-known brands into India to produce electronic hardware for local and export demands. Emphasis on skill development, changed labour policies, rationalisation of the tax structure, better availability of finance, export incentives and others will definitely trigger local production in the near future. We should be ready to absorb the upcoming opportunities by providing near-ideal conditions.

—R.A. Gotor, director - India operations, Hongfa Electroacoustic Co. Ltd

tions that require pure tin-free components.

"Carbon-print resistors are one area which is of immense interest to us. Customers send us description of resistance, wattage, ohm and shape. We can also convert existing SMD resistors to carbon-print ones. In the past, longevity was not good enough but we are able to achieve ten million cycles," said Komiya.

He adds, "Another requirement for changing the materials used is that we need to change or reduce the use of certain materials as per restrictions from the UK due to hazardous materials."

What is next

Cost cutting is an important part of the strategy for components, and in passive components this is led by high technology. "Cost cutting is not as easy as it sounds, as integration is the key area that is being driven

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here. While there have been many incremental changes over the years, the first revolution in this space was that of passives being embedded in PCBs," says Karun Malhotra, director of corporate technology and business development at Murata.

"Overall, there are very limited changes currently being made on the technology side. However, on the production side, there are a lot of changes happening. In the future, one improvement that I expect is that we might have several passive components on a single chip," says Shukla. This will enable designers to reduce the circuit board area consumed and thus deliver a much lower total cost of the product, while at the same time enabling smaller devices.

Some advances also aim to improve things for the production side of things. "Our latest SMD resistors that are currently in development have a special design for automatic surface mounting," according to Komiya. ●

How to convert through-hole resistors into surface-mounts ones

The ZI-form option from TT Electronics allows conversion of axial through-hole resistors into a surface-mount format. This makes these available to designers of SMD circuit resistors with characteristics that are simply not available in flat-chip or MELF styles. For example, power resistors with ratings up to 5W that dissipate heat into the air can now be surface mounted—a valuable alternative to an array of several chip resistors that dump heat into the PCB. Precision metal-film parts are available that give combinations of power rating and stability that are not available in thin-film chips.